

Connect and Protect: Networks and Network Security

Module 1

Network architecture

Get started with the course

Introduction to Course 3

Video • 1 min

Course 3 overview

Reading • 4 min

Helpful resources and tips

Reading • 4 min

Introduction to networks

Welcome to module 1

Video • 29 sec

Chris: My path to cybersecurity

Video • 3 min

What are networks?

Video • 1 min

Tina: Working in network security

Video • 2 min

Emmanuel: Useful skills for network security

Video • 2 min

Network tools

Video • 2 min

Network components, devices, and diagrams

Reading • 4 min

Practice: Design a local area network

Ungraded Plugin • 10 min

Cloud networks

Video • 2 min

Cloud computing and software-defined networks

Reading • 8 min

Identify: Functions of network tools

Ungraded Plugin • 10 min

Test your knowledge: Introduction to networks

Practice Assignment • Grade: 100%

Network communication

Introduction to network communication

Video • 2 min

The TCP/IP model

Video • 2 min

The four layers of the TCP/IP model

Video • 1 min

Learn more about the TCP/IP model

Reading • 4 min

The OSI model

Reading • 8 min

Test your knowledge: Network communication

Practice Assignment • Grade: 100%

Local and wide network communication

IP addresses and network communication

Video • 2 min

Components of network layer communication

Reading • 4 min

Test your knowledge: Local and wide network communication

Practice Assignment • Grade: 100%

Review: Network architecture

Wrap-up

Video • 39 sec

Glossary terms from module 1

Reading • 4 min

Module challenge 1

Graded Assignment • Grade: 100%

Module 2

Network operations

Introduction to network protocols

Welcome to module 2

Video • 1 min

Network protocols

Video • 3 min

Common network protocols

Reading • 4 min

Additional network protocols

Reading • 4 min

Arturo: Working in network security

Video • 2 min

Wireless protocols

Video • 1 min

The evolution of wireless security protocols

Reading • 4 min

Test your knowledge: Recognize different network protocols

Practice Assignment • Grade: 100%

System identification

Firewalls and network security measures

Video • 4 min

Virtual private networks (VPNs)

Video • 2 min

Security zones

Video • 3 min

Subnetting and CIDR

Reading • 4 min

Proxy servers

Video • 3 min

Virtual networks and privacy

Reading • 4 min

VPN protocols: Wireguard and IPsec

Reading • 4 min

Test your knowledge: System identification

Practice Assignment • Grade: 100%

Review: Network operations

Wrap-up

Video • 51 sec

Glossary terms from module 2

Reading • 4 min

Module challenge 2

Graded Assignment • 50 min

Module 3

Secure against network intrusions

Module 4

Security hardening

Glossary terms from module 2

0:01 / 3:04

Terms and definitions from Course 3, Module 2

Address Resolution Protocol (ARP): A network protocol used to determine the MAC address of the next router or device on the path

Cloud-based firewalls: Software firewalls that are hosted by the cloud service provider

Controlled zone: A subnet that protects the internal network from the uncontrolled zone

Domain Name System (DNS): A networking protocol that translates internet domain names into IP addresses

Encapsulation: A process performed by a VPN service that protects your data by wrapping sensitive data in other data packets

Firewall: A network security device that monitors traffic to or from your network

Forward proxy server: A server that regulates and restricts a person's access to the internet

Hypertext Transfer Protocol (HTTP): An application layer protocol that provides a method of communication between clients and website servers

Hypertext Transfer Protocol Secure (HTTPS): A network protocol that provides a secure method of communication between clients and servers

IEEE 802.11 (Wi-Fi): A set of standards that define communication for wireless LANs

Network protocols: A set of rules used by two or more devices on a network to describe the order of delivery of data and the structure of data

Network segmentation: A security technique that divides the network into segments

Port filtering: A firewall function that blocks or allows certain port numbers to limit unwanted communication

Proxy server: A server that fulfills the requests of its clients by forwarding them to other servers

Reverse proxy server: A server that regulates and restricts the internet's access to an internal server

Secure File Transfer Protocol (SFTP): A secure protocol used to transfer files from one device to another over a network

Secure shell (SSH): A security protocol used to create a shell with a remote system

Security zone: A segment of a company's network that protects the internal network from the internet

Simple Network Management Protocol (SNMP): A network protocol used for monitoring and managing devices on a network

Stateful: A class of firewall that keeps track of information passing through it and proactively filters out threats

Stateless: A class of firewall that operates based on predefined rules and does not keep track of information from data packets

Subnetting: The subdivision of a network into logical groups called subnets

Transmission Control Protocol (TCP): An internet communication protocol that allows two devices to form a connection and stream data

Uncontrolled zone: The portion of the network outside the organization

Virtual private network (VPN): A network security service that changes your public IP address and masks your virtual location so that you can keep your data private when you are using a public network like the internet

Wi-Fi Protected Access (WPA): A wireless security protocol for devices to connect to the Internet

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